

Assignment. 1

1. Write a note on cyber security

→ Cyber security refers to the practice of protecting computer, network, servers, mobile device, software and data from unauthorized access, cyber attack, damage, theft, or disruption.

Importance of cyber security:

- i. protects personal data such as password, bank details and identity information.
- ii. prevents unauthorized access to computer and network.
- iii. protects organization from data breach and financial losses.
- iv. prevents malware and virus attacks.

* Common cyber security Measures.

- using strong password and multi-factor authentication.
- installing antivirus and anti-malware software.
- keeping operating systems and applications updated.
- using firewalls.
- encrypting sensitive information.
- Regularly backing up important data.
- Educating users about phishing and other cyber threats.

Cyber security is essential in today's digital world because it protects information, system and users from continuously evolving cyber threats.

2. Explain CIA triad of Information Security.

→ The CIA triad is a fundamental model of information security. CIA stands for

C - Confidentiality

I - Integrity

A - Availability

1. Confidentiality: means ensuring that information is accessible only to authorized users.

Eg: A bank account password should only be accessible to the account holder.

Methods: Password, encryption, access control.

2. Integrity: means ensuring that information remains accurate, complete and unaltered by unauthorized people.

Method: Hashing, digital controls, data validation.

3. Availability: means ensuring that information and systems are accessible to authorized users whenever required.

Example: An online banking website should be available when customers need to access their accounts.

Method: Regular backup, disaster recovery, protection against DDoS attack.

The CIA triad provides the basic foundation for designing and maintaining secure information systems.

3. What are the different types of cyber crimes?

→ Cyber Crime is any illegal activity in which a Computer, network, digital device, or the internet is used as a tool or target.

Major Types of Cyber Crimes

i. Hacking : unauthorized access to a Computer system or network.

ii. Phishing : Attackers send fake emails, message or website to trick users into revealing password, banking information, or other sensitive

iii. Identity Theft : Stealing someone's personal information and using it for fraudulent activities.

iv. Online banking and financial fraud : unauthorized transaction or theft of money through online banking, cards, or digital payment system.

v. Cyber Stalking : using social media, messaging platform, or other online service to threaten, insult, or harass someone.

vi. Cyber Stalking : Repeatedly monitoring, threatening or harassing a person using digital platform.

vii. Ransomware attacks : malware encrypts a victim's file and demands money in exchange for restoring access.

Cyber crimes can cause financial losses, privacy violation, reputational damage.

11. What are the different types of malware?

→ malware stands for malicious software. It is software intentionally designed to damage system, steal information, gain unauthorized access or disrupt.

Different Types of Malware

i. Virus: attaches itself to a legitimate file or program and spreads when the infected file is executed.

ii. Worm: Can replicate itself and spread automatically across network without requiring a user to execute an infected file.

iii. Trojan horse: disguises itself as a legitimate or useful software but performs malicious activities after installation.

iv. Ransomware: encrypts files or locks system and demands payment from the victim.

v. Spyware: secretly monitors user activities and may collect sensitive information such as passwords and browsing data.

vi. Adware: display unwanted advertisement and may sometime track user activity.

vii. Rootkit: Designed to hide malicious software or unauthorized activities deep inside a system.

viii. Keylogger: record keystroke made by a user and can be used to steal password and other sensitive information.